

1. Introducción

One of the motivations of the early developments of thermodynamics was the optimization of heat engines, as steam engines, which transform partially heat into work, and which led to the first industrial revolution between the end of the eighteenth century and the beginning of the nineteenth century. This provides an interesting illustration of how an engineering problem coupled with economical needs stimulated the occurrence of a new fundamental science. Nowadays, we live in a similar situation: we are experiencing an industrial revolution based on information processing and miniaturized engines and, on the other side, we are conscious of the need to incorporate ecological restrictions into the economical progress. To convey some flavour of these problems, we present in this chapter applications of non-equilibrium thermodynamics to heat engines working with a finite non-vanishing rate. This approach is referred to as *finite-time thermodynamics* and received a strong impetus during the last quarter of the twentieth century.

In 1824, Carnot arrived to the conclusion that the efficiency of a heat engine operating between a hot and a cold heat reservoir is maximum in reversible processes, and that it depends only on the temperature of the two reservoirs, but not on the working substance. Here, Carnot's ideas are extended by including considerations not only on the efficiency but also on the power developed by the engine when the cycle is performed in a finite time. This issue is of importance from the economical perspective, because what the society needs is an amount of work in a finite time, rather than work supplied at an infinitesimal small rate, as in reversible engines.

Though the mutually conflicting demands in the maximization of efficiency and power are especially illustrative of the conceptual problems dealt with in this chapter, we also compare other optimization criteria, as for instance, those based on the minimization of entropy production, and on maximization of the power minus the product of temperature and entropy production. This diversity of criteria illustrates the conceptual subtleties arising in finite-time thermodynamics. Furthermore, the time

constraints must be included in realistic models of engines to optimize their design and performance.

Finite-time thermodynamics deals with a very wide collection of situations. In the present chapter, we only present some simple and pedagogical illustrations, based on Carnot's engine, but many other problems have been investigated in finite-time thermodynamics as, for instance, how to optimize a thermodynamic cycle when its total duration is fixed. In this case, the several branches of the cycle may be performed at different rates, and the problem is to find the rates of each process in such a way that some optimization criterion (power output maximization, for instance) is satisfied. Finite-time thermodynamics is not limited to Carnot cycle, but it encompasses many other cycles (Otto, Brayton, Stirling, Diesel, etc.), and it is concerned not only with ideal gases but also with other kinds of fluids, as polytropic fluids. Particular studies have been devoted to refrigerators, heat pumps, engines with heat regeneration, several coupled engines, and have been applied to solar engines, photovoltaic cells, or to the evaluation of the maximum efficiency of the conversion from solar energy to wind energy in the planet. In short, finite-time thermodynamics allows us to understand in a simple way the essential role of time constraints on the thermodynamic cycles. This is a clear illustration of the importance of including non-equilibrium thermodynamics both in the assessment and design of heat engines as well as in the understanding of some natural phenomena.

2. Ciclo de Carnot (aproximación clásica)

Let us briefly recall the essentials of the classical analysis of Carnot's engine. The latter is operating between two heat reservoirs at temperatures T_1 and T_2 ($T_1 > T_2$), it takes an amount of heat Q_1 from the hot one, transforms a part of it into work W , and delivers the remaining heat Q_2 to the cold reservoir. Since the engine operates cyclically, Q_1 , Q_2 , and W refer, respectively, to the amounts of heat, absorbed and delivered, and the work carried out by the engine during one cycle.

The internal energy of the engine does not change during one cycle, because it is a state function, so that the first law takes the form

$$Q_1 = W + Q_2 \quad (1)$$

If we ask about the efficiency η of the cycle – namely the work produced W divided by the heat supplied Q_1 to the engine – one has

$$\eta \equiv \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} \quad (2)$$

Kelvin–Planck formulation of the second law states the impossibility of building a cyclical heat engine, which transforms heat completely into work. Indeed, the maximum value of η is limited by the minimum value of the heat Q_2 delivered to the cold reservoir. Therefore, to evaluate the maximum efficiency, the second law must be taken into account. Accordingly, the global balance of entropy is never negative, i.e.

$$\Delta S_{\text{tot}} = \Delta S_1 + \Delta S_2 + \Delta S_{\text{engine}} \geq 0 \quad (3)$$

where ΔS_1 , ΔS_2 , and ΔS_{engine} are the variations of entropy, respectively, in the reservoirs 1 and 2 and in the engine during one cycle. Since the engine operates cyclically, and recalling that the entropy is a function of state, one has $\Delta S_{\text{engine}} = 0$. Moreover, the temperature of each reservoir remains constant, so that, according to Clausius' definition of entropy, the variations of their respective entropies are $\Delta S_1 = -Q_1/T_1$ and $\Delta S_2 = Q_2/T_2$. Therefore, (3) becomes

$$\Delta S_{\text{tot}} = -\frac{Q_1}{T_1} + \frac{Q_2}{T_2} = \Delta_{\text{irr}} S \geq 0 \quad (4)$$

The equality sign corresponds to reversible processes and $\Delta_{\text{irr}} S$ stands for the entropy produced during one cycle by irreversible processes (as for instance thermal resistance or mechanical and hydrodynamical friction). After combining (2) and (4), it follows that the efficiency obeys the inequality

$$\eta \leq 1 - \frac{T_2}{T_1} \quad (5)$$

This expression shows that the efficiency is maximum when the engine operates reversibly: indeed, in this case the relation (5) becomes an equality, providing the so-called Carnot's efficiency, depending only on the temperature of the sources but not on any detail of the working substance or the engine itself.

However, in order that the engine operates reversibly, one must proceed infinitely slowly, with the result that the power delivered by the reversible engine is in fact zero. This is of no practical interest because, in actual situations, a non-vanishing power is desirable, which implies to obtain a finite amount of work in a finite time. Anyway, one should avoid spending unrealistic efforts to improve engines beyond their actual physical limits; in that respect, the result (5) indicates that a way to enhance the efficiency is to raise as much as possible the temperature of the hot reservoir and to reduce the temperature of the cold reservoir.

A cycle carried out in a finite time means a reduction of the efficiency, and therefore one is faced with an alternative: maximization of efficiency or maximization of power.

3. Ciclo de Carnot en tiempo finito

3.1. Sistemas endorreversibles (modelo de Curzon-Ahlborn)

Curzon and Ahlborn assume that the rate of heat exchanged between the working substance at a temperature T_1' and the heat reservoir at a temperature T_1 is described by Newton's law of heat transport, namely

$$\frac{dQ_1}{dt} = \alpha_1 (T_1 - T_1') \quad (6)$$

where the coefficient α_1 depends on the thermal conductivity of the wall separating the working fluid from the source and on geometrical factors, as the area and the width of the wall. Analogously, the rate of heat delivered to the cold source is

$$\frac{dQ_2}{dt} = \alpha_2 (T_2' - T_2) \quad (7)$$

where α_2 plays a role analogous to α_1 in the heat exchange between the working fluid and the cold reservoir. A sketch of the Curzon–Ahlborn’s model is provided in the figure 1.

It follows from (6) and (7) that the times required for the engine to absorb an amount of heat Q_1 and to deliver the heat Q_2 are, respectively, given by

$$t_1 = \frac{Q_1}{\alpha_1 (T_1 - T_1')} \quad t_2 = \frac{Q_2}{\alpha_2 (T_2' - T_2)} \quad (8)$$

By writing these expressions, we have assumed that the temperatures T_1' and T_2' of the working substance remain constant during the heat exchange process. Furthermore, for the sake of simplicity, it is admitted that the time spent in the adiabatic parts of the cycle is much shorter than the time required for absorbing or delivering the heat. Physically, this implies the absence of frictional losses and inertial effects (e.g. one works with a frictionless piston of zero mass). In this case, the total duration of the cycle is

$$t_{\text{cycle}} \approx t_1 + t_2 \quad (9)$$

This theoretical framework has the advantage of allowing explicit calculations of all the quantities related to the way the heat engine is operating, and to introduce explicitly the notion of time. In comparison with reversible thermodynamics, one makes a step forward by capturing more realistic features of the heat engine.

The power P delivered by the engine during one cycle is the ratio of the work W that is performed and the total duration of the cycle, i.e.

$$P = \frac{W}{t_{\text{cycle}}} \quad (10)$$

Now, we want to obtain both efficiency and power as a function of the duration of the cycle. To do this, we need the expression of the entropy produced during a cycle, which is given by (4). On the other hand, we know that, when an amount of heat Q flows from a system at temperature T to another at temperature T' , the amount of entropy produced is

$$\Delta_{\text{irr}} S = -Q \left(\frac{1}{T} - \frac{1}{T'} \right) \quad (11)$$

The total entropy generated during one cycle is assumed to result exclusively from irreversible heat exchanges between the reservoirs and the working substance, all other possible sources of irreversibility, as for instance viscosity, are neglected for the moment. In the above problem, one has

$$\Delta_{\text{irr}} S = -Q_1 \left(\frac{1}{T_1} - \frac{1}{T_1'} \right) - Q_2 \left(\frac{1}{T_2'} - \frac{1}{T_2} \right) \quad (12)$$

Making use of (4), one obtains

$$-\frac{Q_1}{T_1'} + \frac{Q_2}{T_2'} = 0 \quad (13)$$

Compared to the equivalent result (4), wherein the temperatures T_1 and T_2 of the sources appear, in (13) the temperatures of the working fluid during the isothermal expansion and compression are coming out. The analogy between (13) and (4) for the reversible situation (i.e. with $\Delta_{\text{irr}} S = 0$) underlines that the working fluid itself is behaving in a reversible way, and that the only source of irreversibility is the heat

transfer between systems at different temperatures (i.e. between the working fluid and the corresponding heat reservoirs). This is called an endoreversible heat engine, because irreversibilities find their origin in the contact with the sources but not in internal causes. In particular, for a given compression ratio $V_{\max}/V_{\min}$ of an ideal gas and using the first law of thermodynamics, the values of Q_1 and Q_2 are

$$Q_1 = RT_1' \ln \left(\frac{V_{\max}}{V_{\min}} \right) \quad Q_2 = RT_2' \ln \left(\frac{V_{\max}}{V_{\min}} \right) \quad (14)$$

We now study the conditions under which the power becomes a maximum. The work per cycle W is equal to the difference $Q_1 - Q_2$, and will be the larger when the higher is Q_1 and the lower is Q_2 . According to (13), this requires to increase T_1' and to reduce T_2' , because

$$\frac{Q_1}{Q_2} = \frac{T_1'}{T_2'} \quad (15)$$

But if T_1' increases and T_2' decreases, the rate of heat exchanged with the sources, expressed by (6) and (7), diminishes. Under these circumstances, one has slow cycles with high efficiency but small power: each cycle produces much work but one cycle takes a long time. If the rate of heat is increased by lowering T_1' and rising T_2' , efficiency is reduced: each cycle produces less work but it takes a shorter time. When the rate is too high, the work per cycle becomes very low in such a way that the power decreases and becomes eventually zero. Between these extreme situations with a vanishing power, there is another one with a maximum power output, as sketched in the figure 2. In this figure, the zero values of power output correspond either to very low speed per cycle (high efficiency but small power) or to very high speed per cycle (low efficiency and low power).

To determine the values of T_1' and T_2' corresponding to the maximum power output, at fixed values of the temperatures T_1 and T_2 of the heat reservoirs, let us write explicitly the expression for the power output, namely

$$P(T_1', T_2') = \frac{Q_1 - Q_2}{t_{\text{cycle}}} = \frac{Q_1 - Q_2}{\frac{Q_1}{\alpha_1(T_1 - T_1')} + \frac{Q_2}{\alpha_2(T_2' - T_2)}} \quad (15)$$

Eliminating Q_1 and Q_2 in terms of T_1' and T_2' by using (13), one obtains

$$P(T_1', T_2') = \frac{T_1' - T_2'}{\frac{T_1'}{\alpha_1(T_1 - T_1')} + \frac{T_2'}{\alpha_2(T_2' - T_2)}} \quad (16)$$

After introducing the variables $x_1 \equiv T_1 - T_1'$ and $x_2 \equiv T_2' - T_2$, (16) may be rewritten in the more compact form

$$P(x_1, x_2) = \frac{\alpha_1 \alpha_2 x_1 x_2 (T_1 - T_2 - x_1 - x_2)}{\alpha_2 T_1 x_2 + \alpha_1 T_2 x_1 + (\alpha_1 - \alpha_2) x_1 x_2} \quad (17)$$

The conditions for P being extremum are

$$\frac{\partial P(x_1, x_2)}{\partial x_1} = 0 \quad \frac{\partial P(x_1, x_2)}{\partial x_2} = 0 \quad (18)$$

This leads to two equations for the values of x_1 and x_2 corresponding to the maximum power,

$$\alpha_2 T_1 x_2 (T_1 - T_2 - x_1 - x_2) = x_1 (\alpha_2 T_1 x_2 + \alpha_1 T_2 x_1 + (\alpha_1 - \alpha_2) x_1 x_2) \quad (19)$$

$$\alpha_1 T_2 x_1 (T_1 - T_2 - x_1 - x_2) = x_2 (\alpha_2 T_1 x_2 + \alpha_1 T_2 x_1 + (\alpha_1 - \alpha_2) x_1 x_2) \quad (20)$$

It is verified that

$$x_{2, \max} = \left(\frac{\alpha_1 T_2}{\alpha_2 T_1} \right)^{1/2} x_{1, \max} \quad (21)$$

with

$$x_{1, \max} = \frac{T_1 - (T_1 T_2)^{1/2}}{1 + (\alpha_1 / \alpha_2)^{1/2}} \quad x_{2, \max} = \frac{(T_1 T_2)^{1/2} - T_2}{1 + (\alpha_2 / \alpha_1)^{1/2}} \quad (22)$$

In particular, it is found from (21) and (22) that

$$T_1' = C T_1^{1/2} \quad T_2' = C T_2^{1/2} \quad C \equiv \frac{(\alpha_1 T_1)^{1/2} + (\alpha_2 T_2)^{1/2}}{\alpha_1^{1/2} + \alpha_2^{1/2}} \quad (23)$$

Introducing (22) in (17), we obtain for the maximum power output

$$P_{\max} = \frac{\alpha_1 \alpha_2}{(\alpha_1^{1/2} + \alpha_2^{1/2})^2} (T_1^{1/2} - T_2^{1/2})^2 \quad (24)$$

The efficiency as a function of x_1 and x_2 is

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2'}{T_1'} = 1 - \frac{T_2 + x_2}{T_1 - x_1} \quad (25)$$

Bringing it in the expressions (22) yields the important relation

$$\eta_{\max \text{ power}} = 1 - \left(\frac{T_2}{T_1} \right)^{1/2} \quad (26)$$

It is worth to stress that, in contrast with Carnot's result (5), the efficiency at maximum power output relates to the square root of the ratio of temperatures. Both relations (24) and (26) are of high interest, especially the second one, which is seen to depend only on the values of the temperatures of the reservoirs, in contrast to the first one, which is also function of the heat transfer coefficient introduced in (6) and (7).

As an illustration, consider a power station working, for instance, between heat reservoirs at 565 and 25 °C and having an efficiency of 36%. We want to evaluate this power station from the thermodynamic point of view. It follows from (5) that the Carnot's maximum efficiency is 64.1%; our first opinion on the quality of the power station would be rather negative if Carnot's efficiency is our standard for evaluation. However, according to (26), its efficiency at maximum power is 40%. Thus, if the objective of the power station is to work at maximum power output, it is seen that the efficiency is not bad compared to the 40% efficiency corresponding to this situation. We can therefore conclude that, to have realistic standards for evaluation of an actual heat engine, one needs to go beyond Carnot's efficiency and to incorporate finite-time considerations.

We have stressed that (26) is independent of the heat transfer coefficients, and therefore it is rather general. However, its generality is not so universal as Carnot's result (5), because it has been derived by assuming that thermal conduction is the only source of irreversibility, thus neglecting other possible dissipative processes, and also because it depends on the specific form adopted for the heat transfer law.

3.2. Pérdidas por fricción

In the previous examples, the presence of internal dissipation has been ignored. Dissipation could be, for instance, produced by friction of the piston along the walls, heat leaks due to imperfect insulation of the reservoirs, internal losses in the working fluid, such as turbulence or chemical reactions. A simple way to introduce friction is to assume that the total force acting on the piston is

$$F = pA - f v \quad (29)$$

where $p \times A$ is the force exerted by the gas on the piston of area A and $f \times v$ represents a frictional viscous force, v being the velocity of displacement of the piston and f the friction coefficient. This model was investigated by Rebhan (2002), who assumed that heat transfer effects are arbitrarily fast and that the only source of dissipation is friction. This is the situation analysed below.

During a displacement dx , the work performed by the piston will be

$$dW = p dV - f v^2(t) dt \quad (30)$$

and the work corresponding to one cycle is

$$W = Q_1 - Q_2 - f \int_c v^2(t) dt \quad (31)$$

the integration being performed over the duration of one cycle. The computation of the last term depends on the time dependence of the velocity. Let us qualitatively write

$$W = Q_1 - Q_2 - \frac{4f\delta D^2}{\tau} \quad (32)$$

where $2D$ is the total distance covered by the piston during one cycle, τ is the duration of one cycle, and δ is a fixed numerical parameter characterizing the dependence of the velocity with respect to time (Rebhan 2002). As a result, the power of the engine will be expressed by

$$P = \frac{Q_1 - Q_2}{\tau} - \frac{4f\delta D^2}{\tau^2} \quad (33)$$

For given values of Q_1 and Q_2 , P is a function of τ only. The maximum power is then obtained from the condition that $dP(\tau)/d\tau = 0$, which yields for the optimum time

$$\tau_{\text{opt}} = \frac{8f\delta D^2}{Q_1 - Q_2} \quad (34)$$

Introducing this result into (33), it is found that the maximum power is

$$P_{\text{max}} = \frac{(Q_1 - Q_2)^2}{16f\delta D^2} \quad (35)$$

The efficiency is given by

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} - \frac{4f\delta D^2}{\tau Q_1} \quad (36)$$

Substituting the value for the optimum time, one obtains

$$\eta(\text{max}P) = \frac{1}{2} \left(1 - \frac{Q_2}{Q_1} \right) = \frac{\eta_{\text{Carnot}}}{2} \quad (37)$$

which is of course lower than Carnot's efficiency, but also lower than the Curzon–Ahlborn's result (26) derived for heat losses only.

Just like (26), expression (37) does not depend explicitly on the friction coefficient. Combined effects of heat losses and friction lead to rather cumbersome mathematical relations, which are certainly interesting from a practical perspective, but which do not add essential new ideas to the above considerations.

4. Otros criterios de optimización

Both power and efficiency must therefore be considered in the assessment of actual heat engines. Up to now, we have discussed only two criteria, namely power optimization and efficiency optimization. Other elements may be taken into account to convey the subtleties of the interplay between economical and ecological needs.

One proposal is to minimize the entropy production, which is related to the loss of available work by $W_{\text{lost}} = T_2 \Delta_{\text{irr}} S$; the rate of entropy production over one cycle is, after use is made of expressions (12) for $\Delta_{\text{irr}} S$ and (8) and (9) for t_{cycle} ,

$$\sigma \equiv \frac{\Delta_{\text{irr}} S}{t_{\text{cycle}}} = \frac{Q_1 \left(\frac{1}{T_1'} - \frac{1}{T_1} \right) + Q_2 \left(\frac{1}{T_2} - \frac{1}{T_2'} \right)}{\frac{Q_1}{\alpha_1 (T_1 - T_1')} + \frac{Q_2}{\alpha_2 (T_2' - T_2)}} \quad (39)$$

In terms of x_1 and x_2 defined under (16), this may be written as

$$\sigma = \frac{\alpha_1 \alpha_2}{T_1 T_2} \frac{x_1^2 x_2 T_2 + x_1 x_2^2 T_1}{\alpha_1 T_2 x_1 + \alpha_2 T_1 x_2 + (\alpha_1 - \alpha_2) x_1 x_2} \quad (40)$$

Another ecological criterion, which is often used, is the maximization of the quantity

$$\Psi = \text{power} - T_2 \times \text{entropy production} = P - T_2 \Delta_{\text{irr}} S \quad (41)$$

By doing so, a high power (a factor of economical interest) is promoted, together with low dissipation (a factor of ecological interest), in such a way that one combines both economical and ecological aims. Physically, (41) is related to the useful work available in a process, which is given by $W = W_R - T_2 \Delta_{\text{irr}} S$, W_R being the work performed by the reversible process and $T \Delta_{\text{irr}} S$ the work dissipated as heat by irreversibilities. Dividing both terms of this expression by the cycle duration t_{cycle} gives back (41). From the expressions (17) for the power output and (39) for the rate of entropy production, one obtains

$$\Psi = \frac{\alpha_1 \alpha_2}{T_1} \frac{T_1 (T_1 - T_2) - (T_1 + T_2) x_1 - 2 T_1 x_2}{\alpha_1 T_2 x_1 + \alpha_2 T_1 x_2 + (\alpha_1 - \alpha_2) x_1 x_2} x_1 x_2 \quad (42)$$